Light

GCSE Computer Science - Data Representation: Images and Sound (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

In a digital image, what information does each pixel store?

- **A.** The colour value for that point in the image
- **B.** The file name of the image
- **C.** The date the image was created
- **D.** The camera model used

Q2 (1 mark)

How is digital sound stored in a computer?

- **A.** As a series of binary numbers representing amplitude samples

- ○ **B.** As continuous sound waves on the hard drive
- ○ **C.** As physical vibrations in the memory chips
- ○ **D.** As analogue electrical signals in the processor

Q3 (1 mark)

A 1-bit image can display how many colours?

- ○ **A.** 1 colour
- ○ **B.** 2 colours
- ○ **C.** 8 colours
- ○ **D.** 256 colours

Q4 (1 mark)

What does a higher sample rate allow a digital recording to capture?

- ○ **A.** Higher frequency sounds more accurately
- ○ **B.** Louder sounds
- ○ **C.** Longer recordings
- ○ **D.** More instruments simultaneously

Q5 (1 mark)

Which of these increases the file size of a bitmap image?

- ○ **A.** Reducing the resolution
- ○ **B.** Decreasing the colour depth
- ○ **C.** Increasing the number of pixels
- ○ **D.** Changing the file name

Q6 (1 mark)

An 8-bit sound recording can represent how many different amplitude levels?

- ○ **A.** 8 levels
- ○ **B.** 16 levels
- ○ **C.** 64 levels
- ○ **D.** 256 levels

Q7 (1 mark)

What happens to image quality when resolution is decreased?

- ○ **A.** Quality decreases because there are fewer pixels to show detail
- ○ **B.** Quality improves because the file loads faster
- ○ **C.** Quality stays the same but colours change
- ○ **D.** Quality improves because each pixel is larger

Q8 (1 mark)

If a 10-second audio clip has a certain file size, what would be the file size of a 20-second clip recorded with identical settings?

- ○ **A.** Double the size
- ○ **B.** The same size
- ○ **C.** Half the size
- ○ **D.** Four times the size

Q9 (1 mark)

A greyscale image uses 8-bit colour depth. How many shades of grey can it display?

- ○ **A.** 8 shades
- ○ **B.** 16 shades
- ○ **C.** 128 shades
- ○ **D.** 256 shades

Q10 (1 mark)

What is meant by 'mono' audio?

- ○ **A.** Audio with a single channel of sound data
- ○ **B.** Audio recorded with one microphone but played through two speakers

- ○ **C.** Audio that only plays on the left speaker
- ○ **D.** Audio with no bass frequencies

Q11 (1 mark)

In what order is pixel data typically stored in a bitmap image file?

- ○ **A.** Row by row, from the top-left corner
- ○ **B.** Randomly scattered throughout the file
- ○ **C.** In alphabetical order by colour name
- ○ **D.** Largest colour values first

Q12 (1 mark)

When digital audio is played back, what happens to the stored binary data?

- ○ **A.** It is deleted from storage
- ○ **B.** It is converted to an analogue electrical signal that drives the speakers
- ○ **C.** It remains as binary and speakers read it directly
- ○ **D.** It is transmitted to the internet

Q13 (2 marks)

An image has dimensions of 200 x 150 pixels with 16-bit colour depth. Calculate the file size in bytes.

- ○ **A.** 30,000 bytes
- ○ **B.** 6,000 bytes
- ○ **C.** 480,000 bytes
- ○ **D.** 60,000 bytes

Q14 (2 marks)

Calculate the file size in kilobytes for a 4-second mono audio clip recorded at 11,025 Hz with 8-bit depth. Use 1 KB = 1,000 bytes.

- ○ **A.** 11.025 KB
- ○ **B.** 5.5 KB
- ○ **C.** 352.8 KB
- ○ **D.** 44.1 KB

Q15 (2 marks)

Explain why reducing colour depth from 24-bit to 8-bit would be problematic for a photograph of a sunset.

- ○ **A.** The file would become too large to store
- ○ **B.** The image would become blurry
- ○ **C.** Smooth colour gradients would show visible banding because 256 colours cannot represent subtle transitions between thousands of shades
- ○ **D.** The image would lose its metadata

Q16 (2 marks)

A sound containing frequencies up to 18,000 Hz is sampled at 32,000 Hz. What problem might occur?

- ○ **A.** The recording will be too quiet
- ○ **B.** Frequencies above 16,000 Hz will not be captured accurately and may cause aliasing distortion
- ○ **C.** The file size will be too small
- ○ **D.** The recording will play back too slowly

Q17 (2 marks)

An image is cropped from 1600 x 1200 pixels to 800 x 600 pixels. By what percentage is the file size reduced (assuming colour depth stays the same)?

- ○ **A.** 25% reduction
- ○ **B.** 50% reduction
- ○ **C.** 75% reduction
- ○ **D.** 90% reduction

Q18 (2 marks)

Why is 16-bit audio considered sufficient for most consumer music playback?

- ○ **A.** Because human ears can only hear 16 different sounds
- ○ **B.** Because music only contains 16 different notes

- ○ **C.** Because 16-bit files are small enough to fit on a CD
- ○ **D.** Because 16-bit provides 65,536 amplitude levels, offering a dynamic range of about 96 dB which exceeds typical listening environments

Q19 (2 marks)

A screenshot containing text and simple graphics is saved. Which format would typically produce a smaller file while maintaining perfect quality?

- ○ **A.** JPEG with high compression
- ○ **B.** PNG (lossless compression)
- ○ **C.** Uncompressed BMP
- ○ **D.** JPEG with low compression

Q20 (2 marks)

What is the main difference between WAV and MP3 audio formats?

- ○ **A.** WAV can only store mono audio while MP3 supports stereo
- ○ **B.** WAV files can only be played on Windows computers
- ○ **C.** MP3 has higher quality than WAV
- ○ **D.** WAV stores uncompressed audio data while MP3 uses lossy compression to reduce file size

Q21 (2 marks)

A graphic designer works with images in CMYK colour mode for print. How does this differ from RGB used for screens?

- ○ **A.** CMYK uses more bits per pixel than RGB
- ○ **B.** CMYK uses four colour channels (Cyan, Magenta, Yellow, Black) for printing, while RGB uses three channels (Red, Green, Blue) for screen display
- ○ **C.** CMYK produces brighter colours than RGB
- ○ **D.** RGB can only display 256 colours while CMYK can display millions

Q22 (2 marks)

Professional video production often uses 48,000 Hz audio sample rate instead of the CD standard 44,100 Hz. What is one reason for this?

- ○ **A.** 48,000 Hz produces smaller file sizes
- ○ **B.** 48,000 Hz divides evenly into common video frame rates (24, 25, 30 fps), making audio-video synchronisation easier
- ○ **C.** Human hearing extends to 48,000 Hz
- ○ **D.** 48,000 Hz eliminates the need for bit depth

Q23 (2 marks)

A 1000 x 1000 pixel image with 8-bit greyscale and a 500 x 500 pixel image with 32-bit colour. Which has the larger file size?

- ○ **A.** The greyscale image is larger
- ○ **B.** The colour image is larger

- ○ **C.** They are the same size
- ○ **D.** Cannot be determined

Q24 (3 marks)

A video file stores 30 frames per second, where each frame is a 1280 x 720 pixel image with 24-bit colour.
Analyse the storage requirements for one minute of uncompressed video.

- ○ **A.** Approximately 166 MB - manageable for most devices
- ○ **B.** Approximately 1.66 GB - significant but practical for short clips
- ○ **C.** Approximately 4.98 GB - requiring substantial storage, explaining why video compression is essential
- ○ **D.** Approximately 16.6 GB - impossible to store without compression

Q25 (3 marks)

A musician compares two recordings of the same performance: Recording A (96,000 Hz, 24-bit) and Recording B (44,100 Hz, 16-bit). Analyse the differences in file size and potential quality.

- ○ **A.** Recording A is approximately 1.5x larger with marginally better quality
- ○ **B.** Both recordings would be identical in size and quality
- ○ **C.** Recording B is larger because lower sample rates require more data
- ○ **D.** Recording A is approximately 3.3x larger; it captures frequencies up to 48 kHz and has greater dynamic range, but these benefits may

be inaudible on typical playback equipment

Q26 (3 marks)

A wildlife photographer captures an image of a bird in flight. The camera saves both RAW (14-bit, 50 MB) and JPEG (8-bit, 8 MB) versions simultaneously.
Analyse why a professional would want both versions despite the storage overhead.

- ○ **A.** The JPEG is a backup in case the RAW file becomes corrupted
- ○ **B.** RAW provides maximum editing flexibility with 16,384 brightness levels per channel for exposure correction and colour grading, while JPEG enables immediate sharing and quick preview; professionals archive RAW for future editing possibilities as software improves
- ○ **C.** The JPEG contains different image content than the RAW
- ○ **D.** RAW files cannot be viewed without the JPEG for reference

Q27 (3 marks)

A podcast network distributes episodes as 64 kbps mono MP3 files, while a classical music label releases albums as 24-bit/96 kHz FLAC files.
Analyse why these dramatically different specifications are appropriate for their respective content.

- ○ **A.** The podcast network cannot afford better equipment
- ○ **B.** Podcasts are shorter than albums so need less data
- ○ **C.** Classical music is older so needs more data to preserve it
- ○ **D.** Speech primarily occupies a limited frequency range and tolerates compression well, making 64 kbps adequate; orchestral music contains complex harmonics, wide dynamic range, and subtle

details that benefit from high-resolution lossless preservation where listeners expect audiophile quality

Q28 (3 marks)

When reducing an image from 24-bit colour to 8-bit colour, software offers 'dithering' as an option.
Analyse what dithering does and why it improves the appearance of colour-reduced images.

- ○ **A.** Dithering increases the file size to store more colours
- ○ **B.** Dithering randomly removes pixels to reduce the image size
- ○ **C.** Dithering strategically places pixels of available colours in patterns that the eye blends together, simulating colours that aren't actually present; this reduces visible banding in gradients at the cost of a slightly grainy texture
- ○ **D.** Dithering converts the image to greyscale before reducing colours

Q29 (3 marks)

A recording engineer notices that the loudest parts of a recording show 'clipping' - the waveform appears flat-topped instead of rounded.
Analyse what causes this problem and how recording settings could prevent it.

- ○ **A.** The sample rate was too low, missing the peaks of the waveform
- ○ **B.** When amplitude exceeds the maximum value that can be represented by the bit depth, it is truncated to the maximum, creating flat-topped distortion; recording at lower input levels or using higher bit depth (more headroom) prevents clipping
- ○ **C.** The microphone was too close to the sound source

○ **D.** Clipping is caused by dust on the recording equipment

Q30 (3 marks)

Both image and sound digitisation involve trade-offs between quality and file size.
Analyse how the concept of 'diminishing returns' applies to both media types when increasing quality settings.

○ **A.** There are no diminishing returns - doubling quality always doubles perceptible improvement

○ **B.** Higher settings always produce worse results due to processing overhead

○ **C.** Beyond certain thresholds (approximately 300 DPI for print images, 16-bit/44.1 kHz for consumer audio), further increases in resolution, colour depth, sample rate, or bit depth produce diminishing perceptible improvements while continuing to increase file size linearly or faster; the optimal settings balance perceptible quality against storage, bandwidth, and processing costs

○ **D.** Diminishing returns only apply to sound, not images

Total: **55** marks • Time: **30 min**

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